

Milestone Three Narrative

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Briefly describe the artifact. What is it? When was it created?

The artifact selected for enhancement is a full-stack, data-driven web application dashboard developed for Grazioso Salvare, an international animal rescue organization. Created in 2025 during the CS 340 Client-Server Development course, this application utilizes Python Dash and Plotly to render interactive filtering tables, demographic pie charts, and real-time geospatial coordinate tracking maps. The system allows rescue personnel to dynamically filter for specialized animal populations, such as water-rescue or disaster-tracking dogs, providing crucial data visualizations to support critical organizational decision-making.

Justify the inclusion of the artifact in your ePortfolio. Why did you select this item? What specific components of the artifact showcase your skills and abilities in software development? How was the artifact improved?

This artifact is included in my ePortfolio because it demonstrates my ability to engineer low-level computational logic and manage complex data structures directly within application memory, rather than relying blindly on pre-built framework utilities. To showcase intermediate programming capabilities, the artifact was overhauled and improved in two specific areas:

- **Custom Data Structure Layout Partitioning:** I restructured how query results are held and manipulated in local runtime memory. The dataset is parsed as a sequential Python List structure containing encapsulated Key-Value Hash Maps (Python dictionaries). During execution, the system dynamically spins up secondary left, middle, and right call-stack array partitions to segment memory chunks on the fly.
- **Implementation of an Optimized QuickSort Algorithm:** I replaced the generic, black-box native table sorting operations with a manually coded, recursive QuickSort algorithm

built using a divide-and-conquer processing loop. By selecting a structural pivot point and partitioning memory sub-arrays, the application now reorganizes thousands of data rows under a highly optimized average time complexity of $O(n \log n)$.

Did you meet the course outcomes you planned to meet with this enhancement in Module One? Do you have any updates to your outcome-coverage plans?

This technical enhancement successfully met and advanced the course outcomes originally targeted in Module One:

Outcome 3: Design and evaluate computing solutions... managing design trade-offs. This was achieved by choosing an in-memory, recursive QuickSort routine over a database-side calculation engine. This decision purposefully managed the system-level trade-offs between local memory consumption on the application machine and hardware processing cycles on the external database cluster.

Outcome 4: Demonstrate an ability to use well-founded techniques, skills, and tools. This was demonstrated by implementing clean, algorithmic file modules (QuickSort.py), wiring custom callback parameters within the Dash controller logic, and using defensive programming key-validity fallback techniques (.get()) to ensure stable runtime data rendering.

Reflect on the process of enhancing and modifying the artifact. What did you learn as you were creating it and improving it? What challenges did you face?

The process of modifying and improving this artifact provided deep insights into professional algorithm lifecycle optimization. I learned that data structures and algorithms are completely codependent; an advanced algorithm is useless without structured memory blocks to manipulate. A major technical challenge faced during development was intercepting the frontend framework's native web actions to inject the custom sorting loop without causing rendering latency. Overcoming this roadblock involved restructuring the main application controller callback loops to accept dynamic table header inputs and translate multi-type DataFrame arrays into plain Python list records. This challenge taught me how to trace data mutations across full-stack tiers and write highly performant code, building substantial confidence in my readiness for advanced database and software quality assurance positions.